CHAPTER 07

# Aldehydes and Ketones II: Enolates

## Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

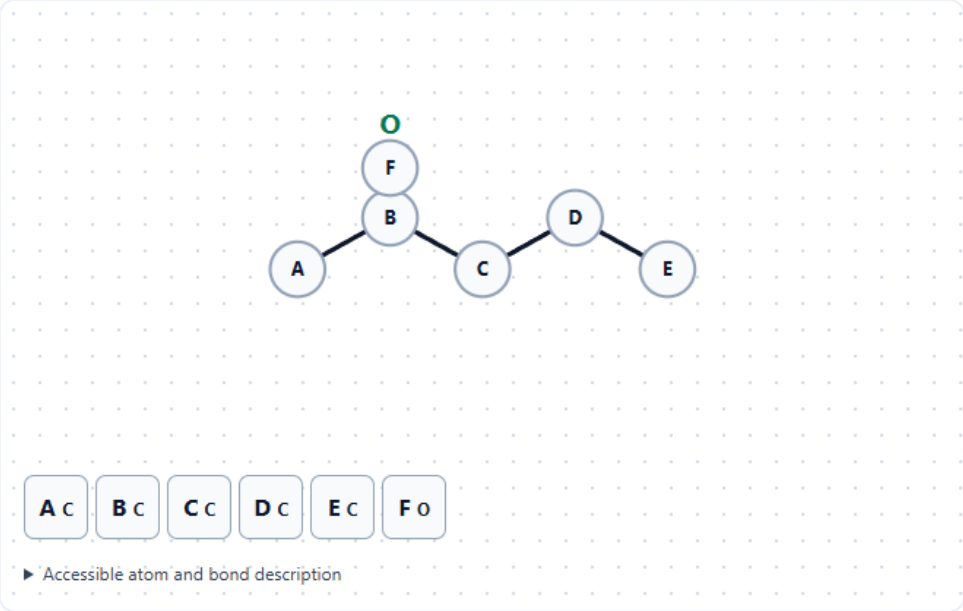
Source snapshot SHA-256: 192b1d0f4620d96577130afa746d201a1d353d725f6eca92a5b974564c627448

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

# alpha-map | Independent exercise

Screenshot: alpha-map-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2]]

## alpha-map:stage-1 | Select both alpha carbons.

Answer type: atoms

## alpha-map:stage-2 | Why are their hydrogens relatively acidic?

Answer type: choice

Option 0 (A): The conjugate base is resonance stabilized

Option 1 (B): They are directly bonded to oxygen

Option 2 (C): All alkane hydrogens have the same acidity

## alpha-map:stage-3 | Enolate resonance contributors differ in…

Answer type: choice

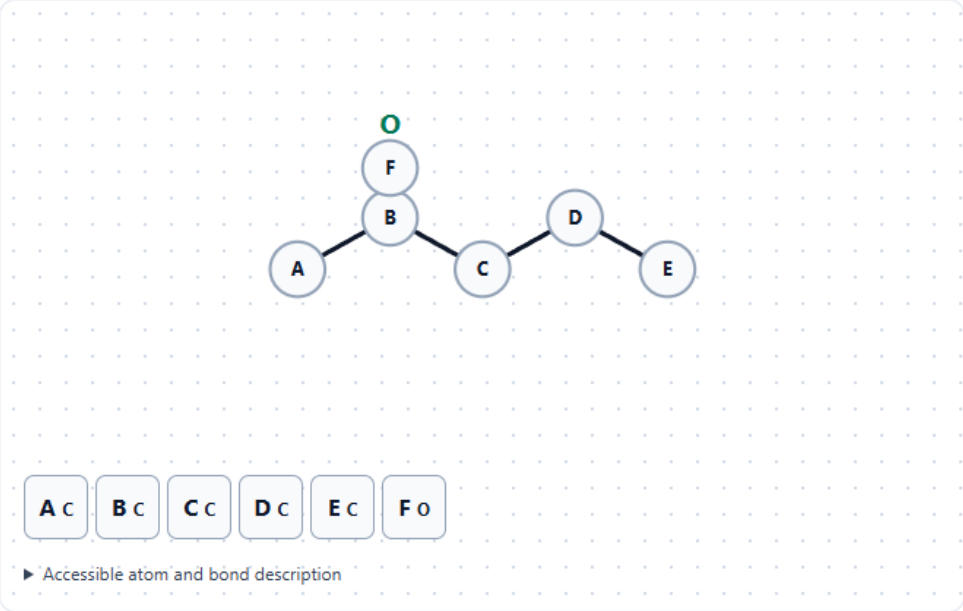
Option 0 (A): Electron placement

Option 1 (B): Location of an H nucleus

Option 2 (C): Molecular formula

# enolate-control | Independent exercise

Screenshot: enolate-control-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2]]

## enolate-control:stage-1 | For typical bulky-base kinetic conditions, select the less hindered alpha carbon.

Answer type: atoms

## enolate-control:stage-2 | Conditions favoring kinetic enolate formation are usually…

Answer type: choice

Option 0 (A): Bulky strong base, low temperature, irreversible deprotonation

Option 1 (B): Long equilibration under reversible conditions

## enolate-control:stage-3 | Neutral keto and enol forms are…

Answer type: choice

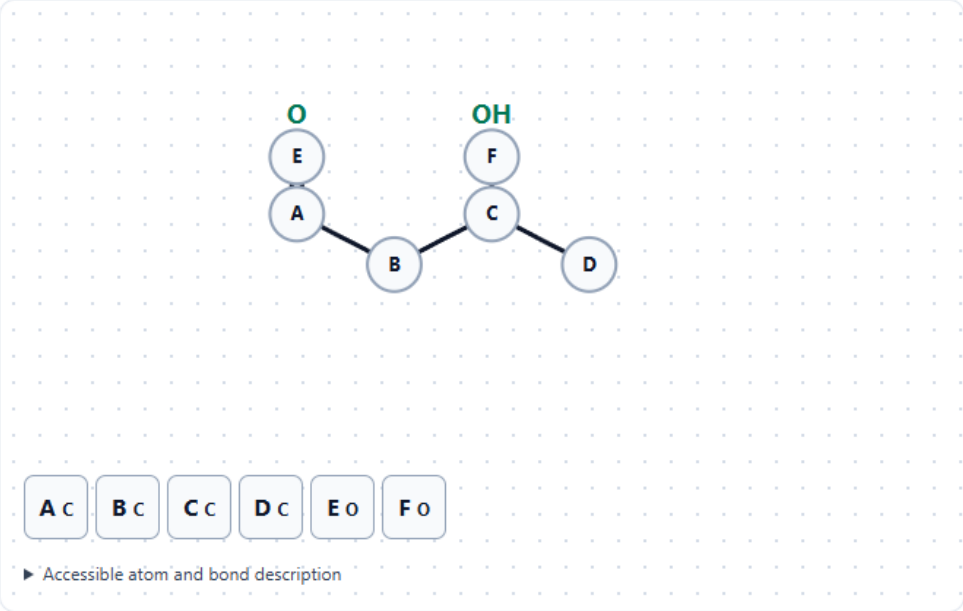
Option 0 (A): Tautomers

Option 1 (B): Resonance contributors

Option 2 (C): Identical conformations

# aldol-map | Independent exercise

Screenshot: aldol-map-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=115.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=115.00; charge=0; explicit H override=none

4 (E): O; x=50.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [0, 4, 2], [2, 5, 1]]

## aldol-map:stage-1 | Select the beta carbon bearing OH.

Answer type: atoms

## aldol-map:stage-2 | The displayed beta-hydroxy product is formed by...

Answer type: choice

Option 0 (A): Aldol addition

Option 1 (B): Aldol dehydration alone

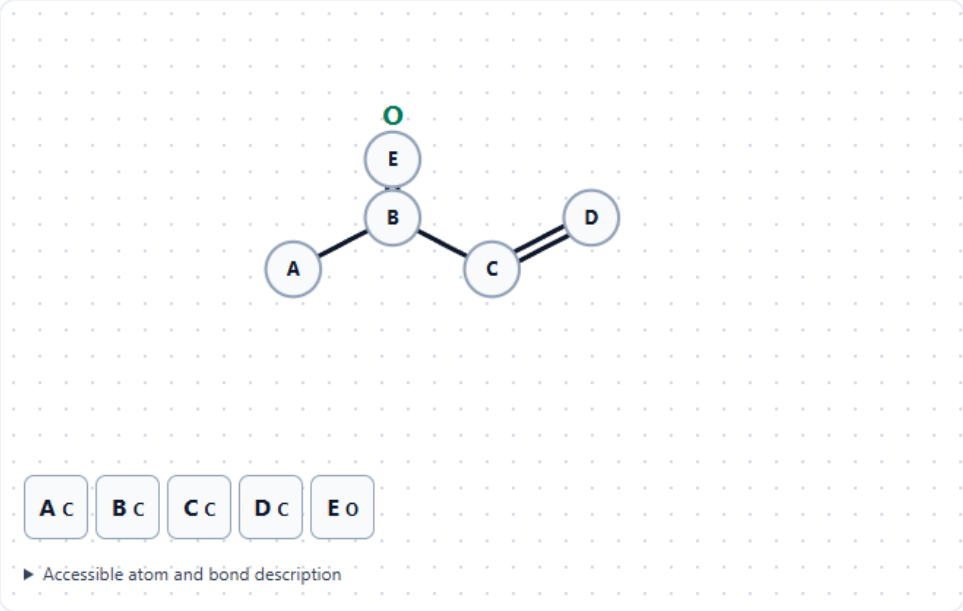
Option 2 (C): Ester hydrolysis

## aldol-map:stage-3 | Select the two carbon endpoints of the bond cleaved in a retro-aldol.

Answer type: atoms

# michael-site | Independent exercise

Screenshot: michael-site-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [1, 4, 2]]

## michael-site:stage-1 | Select the carbonyl carbon.

Answer type: atoms

## michael-site:stage-2 | Select the beta carbon attacked in a Michael addition.

Answer type: atoms

## michael-site:stage-3 | Attack directly at the carbonyl carbon (the C=O site rather than beta) is called…

Answer type: choice

Option 0 (A): 1,2-addition

Option 1 (B): 1,4-addition

Option 2 (C): Retro-aldol